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ABSTRACT

This project analyses the academic performance of university students using data science and machine learning techniques to identify the behavioural and academic factors that most significantly impact their final results. The dataset includes 5,000 student records and 23 features such as attendance rate, exam and project scores, stress levels, study habits, and sleep duration. After thorough data cleaning and preprocessing, several analytical models were implemented: linear and polynomial regression, classification algorithms including decision tree, logistic regression, and random forest, clustering using K-Means, and anomaly detection through both statistical techniques and unsupervised models such as Isolation Forest and Local Outlier Factor. The regression models showed poor performance in explaining final scores, and classification models achieved low accuracy (22%–25%). However, clustering and anomaly detection uncovered valuable insights into student behaviour patterns. The study recommends improving feature engineering, applying more advanced models, and increasing dataset size to enhance prediction accuracy and support more informed academic decision-making.

INTRODUCTION

This project aims to analyse university students' academic performance using data analysis and machine learning techniques. The central issue addressed is the difficulty educational institutions face in accurately predicting student outcomes due to the complex interplay of behavioural, psychological, and academic factors. This challenge is particularly important for institutions that aim to implement early interventions and provide targeted support for students at risk.

The main objective of this study is to uncover meaningful patterns in student behaviour and academic performance that can be used to:

* Predict academic outcomes.
* Classify students into performance categories.
* Detect behavioural anomalies that may signal students in need of support.

To guide this analysis, the following research questions are addressed:

1. To what extent do stress levels, sleep duration, and study hours affect a student's final score?
2. Can a student's grade (A, B, C, D, F) be accurately predicted using academic and behavioural features?
3. Are there observable clusters or behaviour patterns among students based on performance indicators?
4. Are there anomalies in the dataset that may affect model performance or reflect unusual student behaviour?

The dataset consists of 5,000 student records and 23 features, including:

* Attendance rate
* Midterm and final exam scores
* Assignment and project performance
* Participation score

* Study and sleep habits
* Stress level
* Parent education level
* Family income

After thorough data cleaning and preprocessing, the dataset was refined to 4,018 valid records. This process involved handling missing values, correcting gender mislabels, fixing grading inconsistencies, and resolving text formatting issues.

Given the behavioural nature and complexity of the data, machine learning is an appropriate approach for the analysis. The study utilizes:

* Regression models (linear and polynomial) to explore numeric relationships with final scores
* Classification models (logistic regression, decision tree, random forest) to predict letter grades
* Clustering using K-Means to identify student behaviour patterns
* Anomaly detection methods (Isolation Forest, Local Outlier Factor) to flag outliers and abnormal cases

This project presents a complete analytical framework that integrates data preparation, modelling, and evaluation. The insights derived can inform educational policies and enable more personalized, data-driven academic support systems.

METHODOLGY

1. **Data Source and Description**
   Dataset Selection Criteria

Before starting the analysis, the dataset was evaluated against specific criteria to ensure its suitability for data science and machine learning tasks. The following table outlines how the selected dataset meets each requirement:

Criteria	Description	Check (✓/✗)	How the Dataset Meets This Requirement
Topic of Data	The dataset should focus on a meaningful subject to make the analysis more engaging and valuable.	✓	The dataset relates to student performance, including grades, attendance, and activities, allowing for analysis of factors affecting academic success.
Structured Data	The dataset should be in table format, with rows as records and columns as features.	✓	Data is organized in rows and columns, where each row represents a student and each column represents a student's attributes.
Unique Identifier	There should be a column with a unique identifier (e.g., ID, username).	✓	The dataset includes a column named **Student_ID** that holds unique values for each student.
Dataset Size	The dataset should be large enough for meaningful analysis (e.g., 1,000+ rows).	✓	The dataset includes **5,001 rows and 23 columns**, which is sufficient for machine learning analysis.
Numerical Variables	The dataset should include numerical variables (e.g., age, price, score).	✓	The dataset contains 12 numerical variables such as **Attendance (%)**, **Midterm_Score**, **Final_Score**, **Projects_Score**, **Study_Hours**, **Stress_Level**, and **Sleep_Hours**.
Categorical Variables	The data set should include categorical variables (e.g., gender, department, grade).	✓	The dataset includes 7 categorical variables such as **Gender**, **Department**, **Grade**, **Parent_Education_Level**, **Internet_Access_at_Home**, and **Family_Income_Level**.

Table 1: **Dataset Selection Criteria**

The dataset used in this project is titled **"Students Grading Dataset"** and was obtained from Kaggle. The original dataset can be accessed at the following link: https://www.kaggle.com/datasets/omerckn/students-grading-dataset.
This dataset contains academic and behavioral information about **5,000 university students**, covering **23 features**. These features include a combination of academic metrics (e.g., midterm scores, final scores, project scores, attendance), behavioral factors (e.g., stress level, sleep hours, study hours), and demographic variables (e.g., gender, family income, parental education). After preprocessing and cleaning the data (handling missing values, correcting inconsistent labels, and removing duplicates), the dataset was refined to **4,018 valid records** suitable for machine learning tasks.
The dataset contains both **numerical variables** (e.g., Midterm_Score, Final_Score, Attendance, Stress_Level) and **categorical variables** (e.g., Gender, Department, Grade, Parent_Education_Level), enabling a variety of supervised and unsupervised learning techniques. An initial preview using the head () function was performed to inspect the structure and contents of the dataset.

[INFO] Initial Dataset Information:

	Student_ID	First_Name	Last_Name	Email	Gender	Age	Department	Attendance (%)	Midterm_Score	Final_Score	...	Grade	Study_Hours_per_Week	Extracurricular_Activities	Internet_Access_at_Home	Parent_Education_Level	Family_Income_Leve
0	S1000	Omar	Williams	student0universitycom	male	22	Engineering	52.29	55.03	57.82	...	F	6.2	No	Yes	High School	Mediur
2	S1002	Ahmed	Jones	student2universitycom	male	24	Business	57.19	67.05	93.68	...	C	20.7	No	Yes	Masters	Lo
3	S1003	Omar	Williams	student3universitycom	male	24	Mathematics	95.15	47.79	80.63	...	D	24.8	Yes	Yes	High School	Hig
4	S1004	John	Smith	student4universitycom	male	23	CS	54.18	46.59	78.89	...	D	15.4	Yes	Yes	High School	Hig
6	S1006	Ahmed	Jones	student6universitycom	male	24	Business	57.60	66.26	89.07	...	B	21.3	No	Yes	No Education	Lo

5 rows × 25 columns

Figure 1: Preview of the raw dataset before preprocessing

A brief description of some key features is as follows:

- **Final_Score**: Students' performance in exams.
- **Stress_Level (1–10)**: A self-reported indicator of psychological stress.
- **Parent_Education_Level**: Highest level of education attained by a parent.
- **Grade**: The original categorical letter grade assigned to the student (e.g., A, B, C, D, F) before any correction or transformation.

2. Data Cleaning and Preprocessing

A series of preprocessing steps were applied to prepare the data for analysis and ensure high quality:

- **Handling Missing Values:**
  - The Parent_Education_Level column was filled with a new category: "No Education".
  - Rows with missing values in critical columns like Attendance (%) and Assignments_Avg were removed to preserve analysis integrity.

Student_ID	0
First_Name	0
Last_Name	0
Email	0
Gender	0
Age	0
Department	0
Attendance (%)	516
Midterm_Score	0
Final_Score	0
Assignments_Avg	517
Quizzes_Avg	0
Participation_Score	0
Projects_Score	0
Total_Score	0
Grade	0
Study_Hours_per_Week	0
Extracurricular_Activities	0
Internet_Access_at_Home	0
Parent_Education_Level	1794
Family_Income_Level	0
Stress_Level (1-10)	0
Sleep_Hours_per_Night	0

Figure 2: Distribution of missing values across dataset features before handling.

```
datafream.head(5000)
```

	Student_ID	First_Name	Last_N
0	S1000	Omar	Willi
1	S1001	Maria	Br
2	S1002	Ahmed	Jo
3	S1003	Omar	Willi
4	S1004	John	Sr
...	...	...	
4995	S5995	Ahmed	Jo
4996	S5996	Emma	Br
4997	S5997	John	Br
4998	S5998	Sara	D
4999	S5999	Maria	Br

5000 rows × 23 columns

0	S1000	Oma
2	S1002	Ahme
3	S1003	Oma
4	S1004	Joh
6	S1006	Ahme
...	...	
4990	S5990	A
4993	S5993	A
4997	S5997	Joh
4998	S5998	Sar
4999	S5999	Mari

4018 rows × 23 columns

Figure 3: Dataset size before and after removing records with missing critical values

- **Text Cleaning:**
  - A custom clean text () function was applied to remove special characters and excess spaces from textual data.

```
# Function to clean text data
def clean_text(text):
    return re.sub(r'[^A-Za-z0-9أ-ي\s]', '', text) if isinstance(text, str) else text

# Apply text cleaning to all text columns
for col in datafream.select_dtypes(include=['object']).columns:
    datafream[col] = datafream[col].apply(clean_text)

# Display the dataframe after cleaning
datafream.head()
```

Figure 4: Example of textual data before and after applying the clean text () function

Gender Correction:

- Gender labels were validated and corrected based on name-gender matching against predefined male/female name lists.

First_Name	Last_Name	Email	Gender	Corrected_Gender
Omar	Williams	student0universitycom	Female	male
Ahmed	Jones	student2universitycom	Male	male
Omar	Williams	student3universitycom	Female	male
John	Smith	student4universitycom	Female	male
Ahmed	Jones	student6universitycom	Male	male

Figure 5: Correction of mislabelled gender values based on first name comparison

- **Grade Correction:**
    - A custom grading function recalculated letter grades based on Total_Score. Over **3,200 incorrect grade entries** were detected and corrected.

```
[INFO] Students with Incorrect Grades:
     Student_ID  Total_Score Grade Corrected_Grade
2         S1002        70.30     D               C
3         S1003        61.63     A               D
4         S1004        66.13     F               D
6         S1006        83.21     F               B
7         S1007        81.93     F               B
...         ...          ...   ...             ...
4990      S5990        81.18     F               B
4993      S5993        85.86     A               B
4997      S5997        54.25     A               F
4998      S5998        55.84     A               F
4999      S5999        77.86     F               C

[3262 rows x 4 columns]
```

Figure 6: Comparison of original and corrected grade values after validation against total scores.

- **Duplicate Removal:**
    - Duplicate records were identified and removed to maintain data consistency and reliability.

```
Number of duplicate rows: 0    Number of rows after removing duplicates: 4018
```

Figure 7: Handling duplication and ensuring its absence.

These preprocessing steps significantly improved the data quality and ensured more reliable model training and evaluation across all tasks.

3. Model Selection and Justification

In this project, various models were selected and implemented across four main tasks—regression, classification, clustering, and anomaly detection—based on the nature of the target variables and the objectives of the analysis. Each model was chosen carefully with clear justifications to ensure suitability for the dataset and research questions.

Regression Models

To analyze relationships between continuous academic or behavioral features (such as stress level, study hours) and students' final scores (a continuous target variable), the following regression models were implemented:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression (degree = 2)
- Non-Linear Regression

Justification: These models are appropriate for examining how multiple numeric features contribute to variations in final scores. Polynomial and non-linear regression were explored to capture any non-linear relationships missed by linear models.

Classification Models

To predict the Corrected Grade (categorical target with values A–F), the following classifiers were used:

- Decision Tree
- Random Forest
- Logistic Regression
- Gradient Boosting

Justification: Classification models are well-suited for predicting categorical outcomes. Multiple algorithms were tested to compare performance, especially in handling **class imbalance**, where the "F" grade was overrepresented. The use of ensemble models such as Random Forest and Gradient Boosting allowed for better generalization.

Clustering Models

To uncover hidden patterns or behavioral groupings among students without predefined labels, the following unsupervised learning techniques were applied:

- K-Means Clustering (with Elbow Method and PCA for visualization)
- Agglomerative Hierarchical Clustering (with dendrogram analysis)

Justification: Clustering enabled segmentation of students into meaningful groups based on academic and behavioral similarities. PCA was applied for dimensionality reduction to enhance cluster visualization.

Anomaly Detection Models

To identify students with unusual behavior or performance patterns, both statistical and machine learning-based anomaly detection techniques were implemented:

- Z-Score (Statistical)
- IQR and Boxplot (Statistical)
- Isolation Forest (Unsupervised)
- Local Outlier Factor (LOF, Unsupervised)

Justification: These methods were chosen to detect students who deviate significantly from the dataset's normal patterns—whether due to high stress with low performance, or other combinations of outlier behavior.

Final Model Selection:

Across all tasks, multiple models were tested and compared based on performance metrics and interpretability. Final model decisions were made by evaluating results from **5-Fold Cross-Validation**, visualizations (e.g., confusion matrix, clustering scatter plots), and key metrics such as **R^2, RMSE, Accuracy, F1-Score**, and **Silhouette Score**.

4. Training Procedures
All models were trained using a **Train/Test Split** of **80% training and 20% testing**, implemented via train_test_split.
To enhance reliability, **5-Fold Cross-Validation** was applied to all regression and classification models.
Justification: K-Fold cross-validation minimizes variance and ensures that results are not dependent on a single random split.

5. Evaluation Metrics

To ensure a reliable and generalizable evaluation of our models, we adopted a comprehensive assessment approach tailored to each model type. All regression and classification models were trained using an 80/20 train-test split and further validated using **5-Fold Cross-Validation (k=5)**. This technique helped reduce overfitting and ensured that the performance metrics were not dependent on a specific data split, thus increasing the credibility of our results.

For **regression models**, we used three primary metrics:

- **R^2 Score**: Measures the proportion of variance in the final score that can be explained by the input features.
- **Adjusted R^2**: Adjusts the R^2 value based on the number of predictors used, offering a more accurate evaluation for models with multiple features.
- **RMSE (Root Mean Squared Error)**: Provides insight into the average prediction error magnitude, with lower values indicating better model accuracy.

These metrics were selected because they are standard in regression analysis and suitable for continuous target variables like the final score. RMSE and R^2 allow us to assess both the accuracy and explanatory power of our models.

For **classification models**, we used:

- **Accuracy**: To evaluate the overall correctness of predictions.
- **Precision, Recall, and F1-Score**: To assess model performance at the class level, especially under class imbalance.
- **Confusion Matrix**: To visualize the distribution of correct vs. incorrect predictions across all grade classes.
- **ROC Curves**: To analyze sensitivity and specificity, especially useful in multi-class scenarios.

We emphasized **F1-Score** in our interpretation due to the presence of class imbalance—specifically, the overrepresentation of the "F" grade—making it a more reliable indicator than accuracy alone.

For **clustering**, we evaluated performance using the **Silhouette Score**, which measures how well-separated and cohesive each cluster is. A higher silhouette score indicates better-defined clusters.

Overall, the selected evaluation metrics were chosen carefully based on the type of prediction task (regression, classification, clustering) and the structure of the dataset (continuous vs. categorical variables, class imbalance). This multi-metric strategy ensured a thorough and fair comparison of model performance.

6. Tools and Libraries
This project was developed using **Python** and relied on the following libraries:

- pandas, numpy – for data manipulation
- matplotlib, seaborn – for data visualization
- scikit-learn – for modeling and evaluation.
- re – for text preprocessing
- Pipeline, KMeans, RandomForestClassifier, SVC, VotingClassifier, and others – for model implementation.

RESULTS AND DISCUSSION

1.Regression Models

All regression models were trained using both an 80/20 train-test split and 5-Fold Cross-Validation (k=5) to ensure consistency and generalizability of the results. Both methods produced nearly identical performance metrics, which indicate reliable evaluation. The target variable in all regression models is Final Score.

1. Simple Linear Regression

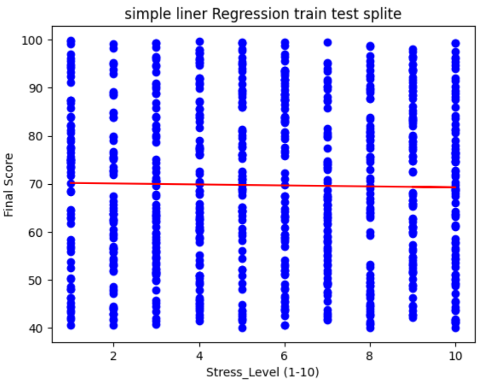


Figure8: Simple Linear Regression

• Research Question: Can the Final Score be predicted based on the Stress Level?
• Independent Variable: Stress Level
• Dependent Variable: Final Score
• Library Used: Linear Regression from sklearn.linear_model
• Performance Metrics:
• R^2 = -0.00014
• Adjusted R^2 = -0.0002
• RMSE = 17.28
• Insights:
• The model does not show a meaningful linear relationship between stress level and final score.
• It is not suitable for prediction in this context, as the model's explanatory power is nearly zero.
• Conclusion: There is no clear relationship between stress level and final score based on this model. This can be seen clearly in the image Figure1: Simple Linear Regression.

Graph Explanation:

Blue Dots:

Each dot represents a student. The position of the dot shows their stress level (horizontal axis) and their grade (vertical axis).

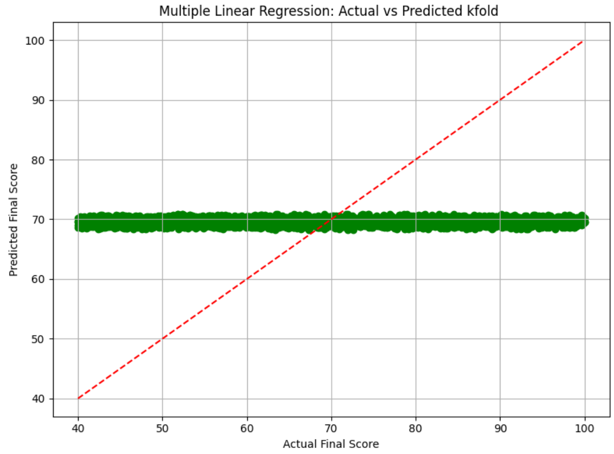
Red Line:

This is the regression line, which represents the best linear fit between stress level and final grade.

Important Note:

The line is nearly horizontal, indicating that the model finds no strong relationship between stress and final grade.

2. Multiple Linear Regression



0-1

Figure9: Multiple Linear Regression

• Research Question: Can the Final Score be predicted using multiple academic and lifestyle factors?
• Independent Variables: Study Hours per Week, Sleep Hours per Night, Stress Level, Attendance Percentage
• Dependent Variable: Final Score
• Library Used: Linear Regression from sklearn.linear_model
• Performance Metrics:
• R^2 = -0.007
• Adjusted R^2 = -0.012
• RMSE = 17.336
• Insights:
• The model explains only a very small portion of the variance.
• There is no significant linear relationship between these combined features and the final score.
• Conclusion: Not a useful predictive model in this case This can be seen clearly in the image Figure2: Multiple Linear Regression.

Graph Explanation:

Green Dots:

Each dot represents a student and shows the difference between the actual and predicted grades.

Red Dashed Line:

This line represents the ideal prediction line. If the model is accurate, all points should be close to this line.

Note:

The points are mostly clustered around the value of 70, indicating that the model tends to ignore.
the actual values and gives nearly constant predictions.

3. Simple Non-Linear Regression

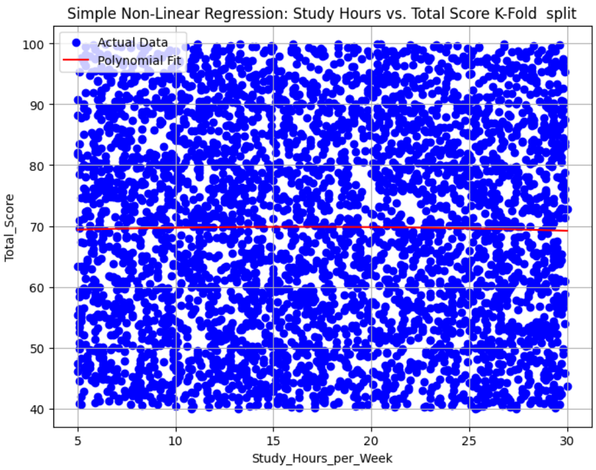


Figure10: Simple Non-Linear Regression

• Research Question: Can the Final Score be predicted based on a non-linear relationship with study hours?
• Independent Variable: Study Hours per Week
• Dependent Variable: Final Score
• Library Used: make pipeline and Polynomial Features from sklearn.preprocessing
• Performance Metrics:
• R^2 = -0.0003
• Adjusted R^2 = -0.0028
• RMSE = 17.1832
• Insights:
• No strong non-linear relationship was detected between study hours and final score.
• Conclusion: The model did not improve over the linear version. This can be seen clearly in the image Figure3: Simple Non-Linear Regression.
• Graph Explanation:
Blue Dots:
Each dot represents a student. The horizontal axis shows the number of study hours per week, and the vertical axis shows the final score.
Red Line:
This is the polynomial regression line, which tries to model a non-linear relationship between study hours and the final score.
Note:
The line is nearly flat, suggesting that the model did not find a strong relationship between study hours and final scores, and is giving nearly constant predictions.

4. Multiple Non-Linear Regression

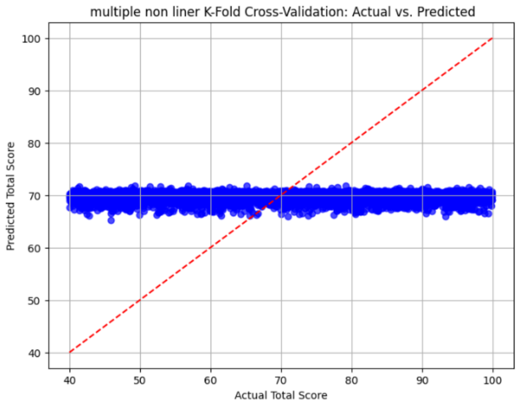


Figure11: Multiple Non-Linear Regression

• Research Question: Can the Final Score be predicted using a non-linear combination of lifestyle and academic features?
• Independent Variables: Age, Attendance, Study Hours per Week
• Dependent Variable: Final Score
• Library Used: make_pipeline and PolynomialFeatures from sklearn.preprocessing
• Performance Metrics:
• $R^2 = 0.0026$
• Adjusted $R^2 = -0.0008$
• RMSE = 17.0872
• Insights:
• The model shows a slight improvement over the linear models.
• However, the relationship is still weak, and the model has limited practical predictive power.
• Conclusion: Only a very weak non-linear relationship exists. This can be seen clearly in the image Figure4: Multiple Non-Linear Regression.
• Graph Explanation:
Blue Dots:
Each dot represents a student. The horizontal position shows the actual grade, and the vertical position shows the predicted grade from the model.
Red Dashed Line:
This is the ideal prediction line (where predicted = actual). If the model were accurate, the dots would be close to this line.
Note:
Most points are clustered horizontally around a score of 70, indicating that the model is almost ignoring the actual values and predicting the same score for all students.

Model	R^2	Adjusted R^2	RMSE
Simple Linear Regression	-0.0001	-0.0002	17.28
Multiple Linear Regression	-0.007	-0.012	17.336
Simple Non-Linear	-0.0003	-0.0028	17.183
Multiple Non-Linear	0.0026	-0.0008	17.087

Table 2: Regression Model Comparison

Model Comparison

Using Table 1: Regression Model Comparison

- The Simple Linear Regression model relied solely on Stress Level as the predictor. It performed poorly, with an R^2 of -0.0001 and an RMSE of 17.28. The regression line was almost flat, indicating no meaningful relationship between stress and final score.
- The Multiple Linear Regression model included multiple academic and lifestyle factors such as Study Hours, Sleep Hours, Stress Level, and Attendance. However, it showed even weaker results with an R^2 of -0.007 and an RMSE of 17.336, suggesting the additional features did not contribute to prediction accuracy.
- The Simple Non-Linear Regression used polynomial transformation on Study Hours, aiming to capture non-linear patterns. While its RMSE improved slightly to 17.183, the R^2 remained low at -0.0003, indicating that the non-linear model still failed to uncover meaningful relationships.
- The Multiple Non-Linear Regression Combined Age, Attendance, and Study Hours with polynomial features. It achieved the best metrics among all models with an R^2 of 0.0026 and the lowest RMSE of 17.087. Although the improvement was marginal, it showed a slight advantage in modeling performance.

Best Model

Multiple Non-Linear Regression was the best-performing model.
Despite its limited accuracy, it outperformed the other models in both error reduction and variance explanation. Further improvements could be achieved by incorporating more relevant features and exploring advanced modeling techniques.

Limitations and Suggestions

Limitations:

- The models showed very weak predictive power, with R^2 values near zero or negative.
- Most predictions clustered around the average (~70), indicating the models failed to capture variance.
- Important influencing features might be missing from the dataset.

Suggestions:

- Incorporate additional features such as student motivation, participation, or quiz scores.
- Try more advanced models like XGBoost or neural networks.

- Use feature transformation (e.g., logarithmic or polynomial) to enhance relationships.

2.Classification Analysis

All classification models were evaluated using both an 80/20 train-test split and 5-Fold Cross-Validation (k=5) to ensure reliable and generalizable results. The goal was to predict the Corrected Grade (A–F) of students using key academic and personal features: Project Score, Midterm Score, Attendance, Study Hours per Week, Sleep Hours per Night, Stress Level, and Final Score. These features were used consistently across all models—Logistic Regression, Random Forest, Decision Tree, and Gradient Boosting—to ensure a fair comparison.

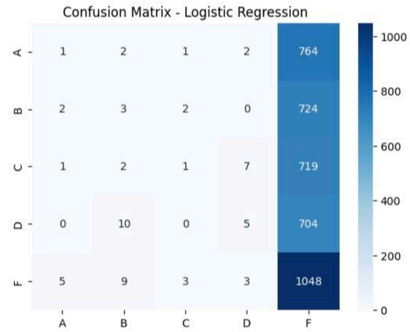
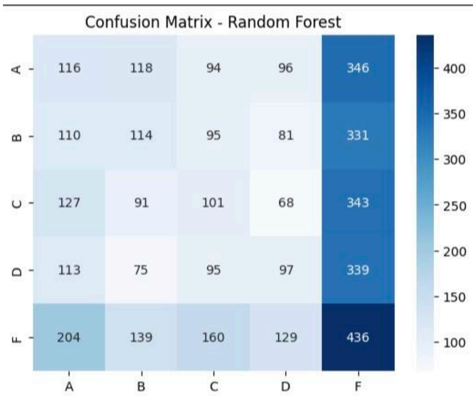


Figure12: Confusion Matrix Logistic Regression

Accuracy: 0.26
Precision: 0.18
Recall: 0.20
F1 Score: 0.09
Interpretation:
This model achieved the highest accuracy. It predicted the failing grade (F) with very high recall (0.98) but had extremely low performance for classes A–D. This indicates a strong bias toward the dominant class. This can be seen clearly in image Figure11**:** Confusion Matrix Logistic Regression.
Usefulness and Insight:
Logistic Regression is useful when identifying students at risk of failing. However, it fails to provide meaningful predictions for higher grades. Regularization and class weighting are



needed.

Figure13: Confusion Matrix Random Forest Classifier

Accuracy: 0.22
Precision: 0.20
Recall: 0.20
F1 Score: 0.19
Interpretation:
Random Forest slightly improved recall for class F (0.41), but predictions for other classes remained weak. The model benefits from ensemble learning but still fails to handle imbalanced classes effectively. This can be seen clearly in the image Figure12: Confusion Matrix Random Forest Classifier.

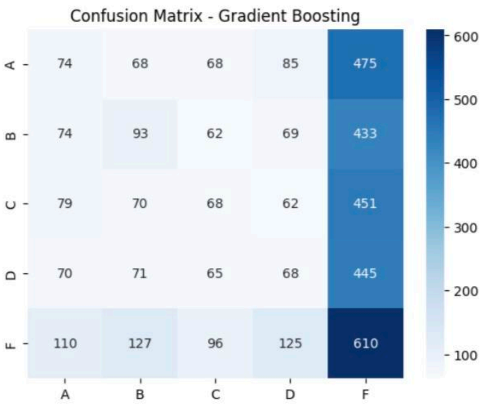
Usefulness and Insight:
It provides slightly better generalization than the Decision Tree. It can be improved with class balancing techniques and parameter tuning. It is moderately useful.

Figure14: Confusion Matrix Decision Tree Classifier

Accuracy: 0.21
Precision: 0.21
Recall: 0.21
F1 Score: 0.21
Interpretation:
This model showed poor performance across all metrics. Class F had slightly better performance due to its larger representation (1068 samples), but classes A–D were predicted poorly. The model appears to suffer from underfitting and lacks complexity to learn deeper patterns This can be seen clearly in the image Figure13: Confusion Matrix Decision Tree Classifier.

Usefulness and Insight:

The Decision Tree was not effective in capturing the relationship between features and grades. It needs hyperparameter tuning to improve depth and structure. It is the least



useful among the models.

Figure15: Confusion Matrix Gradient Boosting Classifier

Accuracy: 0.23
Precision: 0.20
Recall: 0.20
F1 Score: 0.18

Interpretation:
This model performed better than Decision Tree and Random Forest in terms of balance. Recall for class F improved to 0.57. Other classes remained low, but the overall F1 score was more acceptable This can be seen clearly in the image Figure14: Confusion Matrix Gradient Boosting Classifier.

Usefulness and Insight:
Gradient Boosting is promising for multi-class classification. With tuning of learning rate, depth, and estimators, its performance can further improve. It offers the best balance among models.

Model	*Accuracy*	*Precision*	*Recall*	*F1 Score*
Logistic Regression	0.26	0.18	0.20	0.09
Random Forest	0.22	0.20	0.20	0.19
Decision Tree	0.21	0.21	0.21	0.21
Gradient Boosting	0.23	0.20	0.20	0.18

Table 3: Classification Model Comparison

Model Comparisons

Using Table 2: Classification Model Comparison

- Logistic Regression had the highest accuracy (0.26) but was heavily biased toward the F class. It successfully identified failing students but could not predict higher-performing grades (A–D), making it suitable only for detecting at-risk students.
- Random Forest showed moderate performance, with slightly better recall for F (0.41) but continued weakness in other classes. It provides better generalization than Decision Tree but still struggles with class imbalance.
- Decision Tree was the least effective model, with poor results overall. It failed to learn complex relationships between features and grades, likely due to underfitting. It requires deeper structure and parameter tuning to be competitive.
- Gradient Boosting achieved the most balanced performance. With a recall of 0.57 for class F and decent F1 score, it outperformed the other models in stability and class-level fairness. It handled imbalance better and showed potential for further improvement.

Best Model

Gradient Boosting is the best model for this classification task.
It combines moderate accuracy with improved balance across grade categories, making it more reliable for real-world educational analysis. With additional tuning (learning rate, estimators, depth) and class balancing techniques (e.g., SMOTE), its performance can be significantly enhanced.

Limitations and Suggestions

Limitations:

- Class F was overrepresented, causing strong bias in most models.
- Very low performance in predicting higher grades (A–D).
- Class imbalance negatively impacted precision and recall.

Suggestions:

- Apply class balancing techniques such as SMOTE or undersampling.
- Tune model hyperparameters using GridSearchCV or RandomizedSearchCV.
- Try other classification algorithms such as SVM, XGBoost, or LightGBM.

4.Clustering Analysis
Research Question
Can students be grouped into meaningful clusters based on their academic and behavioral features to uncover hidden patterns?
All features in the dataset were used in the analysis, including Final Score, Midterm Score, Project Score, Study Hours per Week, Sleep Hours, Stress Level, and Attendance.

Libraries Used
• sklearn.cluster
• sklearn.decomposition
• scipy.cluster.hierarchy
• matplotlib
• seaborn

Determining the Optimal Number of Clusters

Elbow Method
The Elbow Method shows the Within-Cluster Sum of Squares (WCSS) for different values of k. A visible "elbow" appears at **k = 2**, indicating this is the most efficient number of clusters. After k = 2, the reduction in WCSS becomes marginal, suggesting that increasing k further adds complexity without significant benefit This can be seen clearly in the image Figure16: Elbow Method for optimal k.

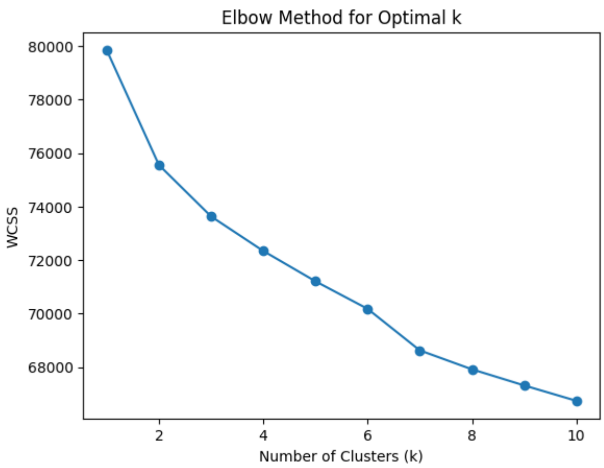


Figure16: Elbow Method for optimal k

Dendrogram (Hierarchical Clustering)
The hierarchical clustering dendrogram proposes **k = 9 clusters** by cutting the tree at a distance of 25. However, this result is purely structural and does not necessarily produce meaningful or interpretable clusters when visualized. This can be seen clearly in image Figure17: Hierarchical Agglomerative Clustering Dendrogram.

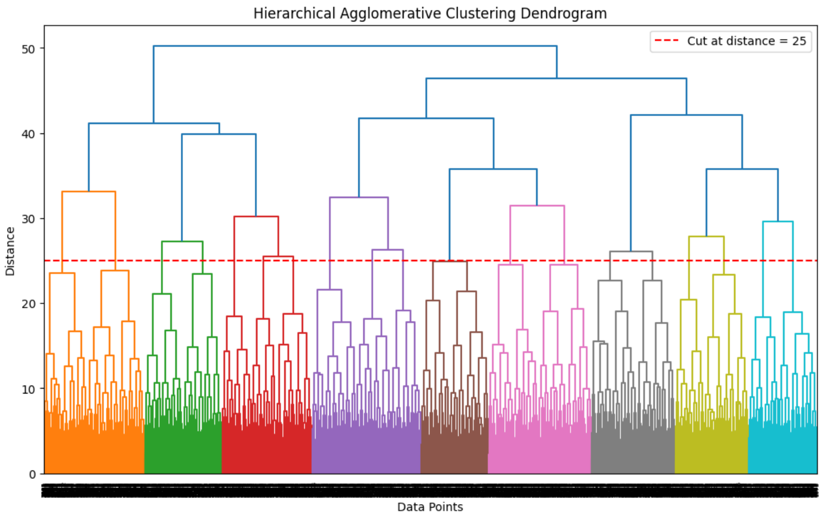


Figure17: Hierarchical Agglomerative Clustering Dendrogram

K-Means Clustering Results and Interpretation

K-Means was applied with different k values and evaluated visually using PCA (Principal Component Analysis) to reduce the feature space to two dimensions.

K-Means Clustering with PCA (k = 2)
This configuration shows two clearly separated clusters. The red stars mark the centroids of each group. This result is consistent with the Elbow Method and provides the most intuitive grouping of students. The clusters appear well-balanced and interpretable. This can be seen clearly in the image Figure18: K-Means Clustering Results with PCA (k = 2)
Silhouette **Score**: 0.063
Insight: Best performance among all tested k-values; supports Elbow Method findings.
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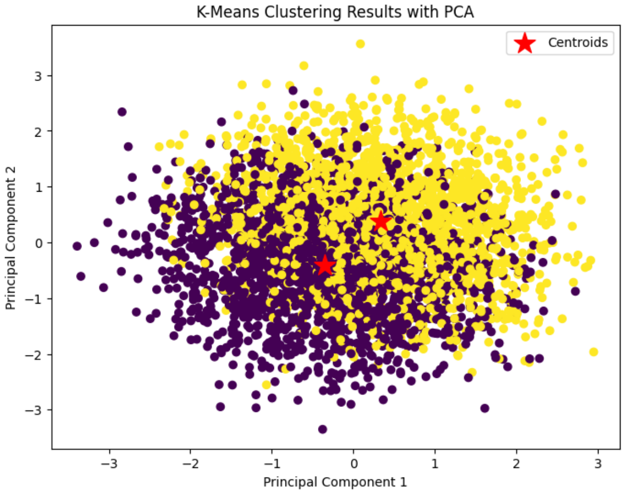

Figure18: K-Means Clustering Results with PCA (k = 2)

K-Means Clustering with PCA (k = 7)
At k = 7, the clusters are heavily overlapping with weak boundaries. The centroids are very close together, indicating that the model is splitting the data without a strong natural distinction. This can be seen clearly in the image Figure19: K-Means Clustering Results with PCA (k = 7)
Silhouette Score: 0.054
Insight: Clusters lacked strong differentiation; centroids were very close.
.

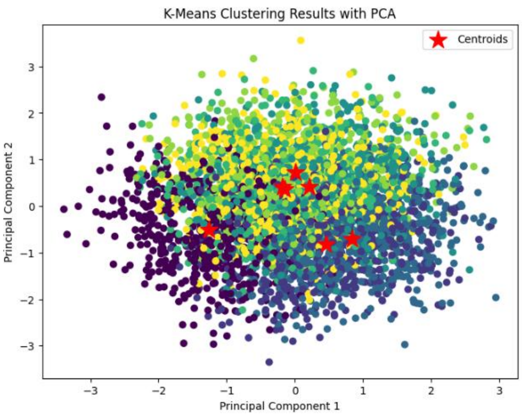

Figure19: K-Means Clustering Results with PCA (k = 7)

- K-Means Clustering with PCA (k = 9)
  Following the suggestion from the dendrogram, the result is visually scattered and noisy. The clusters lack structure and meaningful separation, which leads to over-fragmentation and poor interpretability. This can be seen clearly in the image Figure20: K-Means Clustering Results with PCA (k = 9)

Silhouette Score: 0.055

Insight: Confirmed the visual limitation of dendrogram suggestion.

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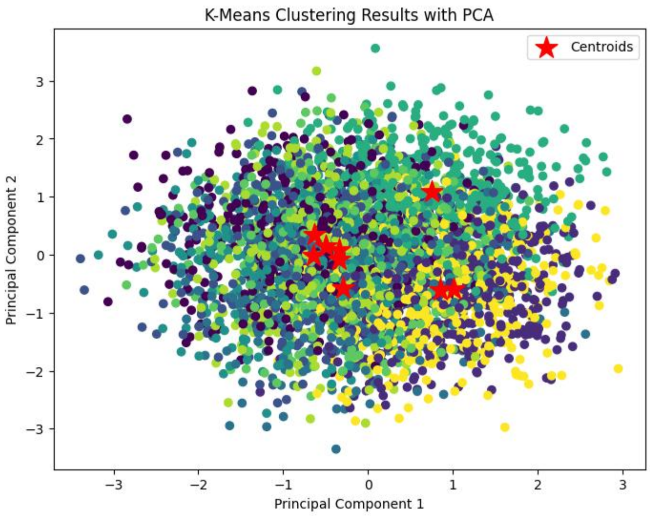

Figure20: K-Means Clustering Results with PCA (k = 9)

Scatter plot Clustering Visualization

This scatter plot shows the hierarchical clustering result based only on Midterm and Final scores. Although it creates four color-coded clusters, they are highly overlapping and random, reinforcing that hierarchical clustering does not produce clear segmentations in this dataset This can be seen clearly in the image Figure21: Scatter plot K-Means Clustering Results based on Student Score

Silhouette Score: 0.055

Insight: Using limited features further weakened clustering performance.

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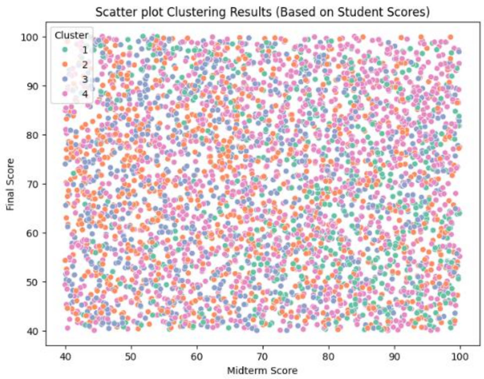


Figure21: Scatter plot K-Means Clustering Results based **on Student Score**

Visualization Insights

- Without PCA: Cluster plots based only on two original features showed unclear and overlapping groupings. This can be seen clearly in the image Figure21: Scatter plot K-Means Clustering Results based on Student Score.

- With PCA: The data becomes easier to interpret in 2D. Clear separation was achieved especially when k = 2. This can be seen clearly in the image Figure18: K-Means Clustering Results with PCA (k = 2).

- PCA enabled the inclusion of multiple features while maintaining clarity in visualization.

Conclusion

The best clustering configuration was K-Means with k = 2 and PCA This can be seen clearly in image Figure18: K-Means Clustering Results with PCA (k = 2).
It provided the clearest and most meaningful separation of students into distinct groups. Other configurations (k = 7, 9) introduced noise, ambiguity, and fragmentation.
Clustering helped reveal hidden student profiles, which can be useful for:

- Designing customized learning interventions
- Targeting academic support programs
- Identifying students with similar academic behaviors

Limitations and Suggestions
Limitations:

- The number of clusters (k) was selected using visual methods (Elbow, Dendrogram) without quantitative validation.
- PCA might have masked some non-linear relationships.
- Clusters at k = 7 and 9 were highly overlapping and hard to interpret.

Suggestions:

- Use quantitative metrics like Silhouette Score or Davies–Bouldin Index for validation.
- Try other clustering algorithms such as DBSCAN or Agglomerative Clustering.
- Add more behavioral features to enrich clustering quality.

4.Anomaly Detection
1. Statistical Methods Results

- **Z-Score:**
  - No anomalies were detected using the Z-Score method.
  - All values of Stress_Level (1-10) and Total_Score were within ±3 standard deviations.
  - This indicates the absence of statistically extreme outliers under the normal distribution assumption.
- **IQR Method:**
  - Also, detected 0 anomalies.
  - Both features fell within the calculated interquartile range limits, showing statistical balance.
  - This confirms that the dataset is statistically stable in terms of these two **features.**
- **Boxplot Visualization:**
  - The Total_Score boxplot (see figure below) shows a symmetric distribution with no outliers, supporting the statistical findings.

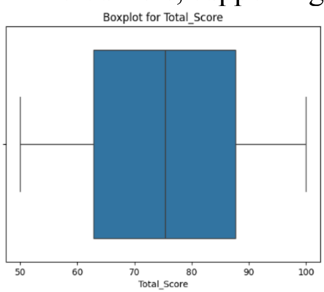
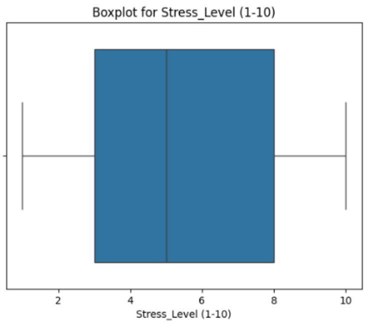


Figure22: Boxplot for total Score

  - The Stress_Level boxplot (see figure below) shows a wide distribution, but no strong visible outliers.



0-1

Figure23: Boxplot for Stress level (1-10)

2. ML Model Results

- **Isolation Forest:**
    - Detected 250 anomalies.
    - These included edge cases such as:
        - Students with high scores and very high stress (possible overachievers under pressure).
        - Students with both low scores and low stress (possibly disengaged).
    - The scatter plot below highlights anomalies (orange) in a two-dimensional view

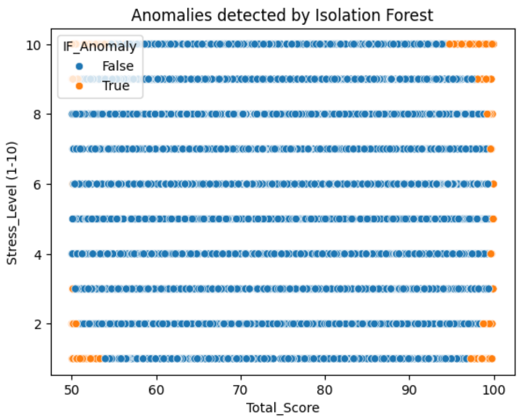


Figure24: Scatter Plot of Anomalies Detected by Isolation Forest

- **Local Outlier Factor (LOF):**
  - Also identified 250 anomalies.
  - Focused on local density variations, revealing context-based outliers that are not extremely in value but deviate from their local neighborhood.

Method	*Anomalies Detected*
Z-Score	0
IQR	0
Isolation Forest	250
Local Outlier Factor (LOF)	250

Table 4: Anomaly Detection Comparison

3. Comparison of Methods
Using Table 3: Anomaly Detection Comparison

- Statistical methods (Z-Score and IQR) failed to detect any anomalies.
- Unsupervised ML models (Isolation Forest and LOF) succeeded in capturing complex multivariate anomalies.
- The fact that both models detected the same number of anomalies (250) strengthens confidence in their robustness and reliability.

4. Interpretation of Anomalies

- Students with high Total_Score and high Stress_Level may reflect overperformance under pressure.
- Students with low stress but also low scores might indicate disengagement or lack of motivation.
- These findings offer meaningful insights into student behavior that simple averages or grades alone cannot reveal.

Limitations and Suggestions
Limitations:

- Statistical methods (Z-Score, IQR) failed to detect any anomalies.
- No labeled anomalies were available to validate model accuracy.
- Some detected anomalies might reflect legitimate edge cases rather than errors.

Suggestions:

- Use datasets with labeled anomalies to evaluate performance more effectively.
- Explore time-based anomaly detection if temporal data is available.
- Try additional models such as One-Class SVM, Autoencoders, or LOF with different distance metrics.

CONCLUSION

This project aimed to analyse student academic performance by applying various data analysis and machine learning techniques, including regression, classification, clustering, and anomaly detection. The dataset included key academic and behavioural features such as grades, study habits, sleep hours, stress levels, and attendance.

Regression models, including linear and non-linear approaches, performed poorly overall, with R^2 values close to zero, indicating a lack of strong predictive relationships. The best regression model was the Multiple Non-Linear Regression, though it showed only a slight improvement.

In classification, four models were tested to predict corrected grades. While Logistic Regression achieved the highest accuracy, it was heavily biased toward the failing class. Gradient Boosting provided the most balanced results and was considered the best model for multi-class prediction.

Clustering analysis revealed that K-Means with $k = 2$ using PCA offered the clearest and most interpretable grouping of students, while higher values of k led to over-fragmentation. Anomaly detection using Isolation Forest and LOF identified 250 potential outliers, offering insights into students with unusual performance or behaviour.

Challenges included class imbalance, limited correlation between features and outcomes, and lack of labelled data for anomaly validation. Evaluation relied on metrics such as RMSE, R^2, accuracy, precision, recall, and F1-score, supported by visual tools including confusion matrices, elbow plots, scatter plots, and dendrograms.

To improve future analyses, we suggest using more advanced algorithms like XGBoost or SVM, applying class balancing methods, engineering new features, and expanding the dataset. Additionally, integrating temporal data and cross-source variables could enhance model accuracy and practical insights.

Overall, this analysis provides a foundation for developing personalized learning strategies, identifying at-risk students, and improving academic support systems through data-driven decisions.

REFERENCES

Kaggle. (n.d.). *Students Grading Dataset*. Retrieved from https://www.kaggle.com/datasets/omerckn/students-grading-dataset
This dataset was used as the primary data source for all analyses in the project.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). *Scikit-learn: Machine learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.

This paper supported the usage of machine learning models and evaluation metrics implemented in the project (Pedregosa et al., 2011).

Raschka, S., & Mirjalili, V. (2020). *Python machine learning: Machine learning and deep learning with Python, scikit-learn, and TensorFlow 2* (3rd ed.). Packt Publishing.

This textbook supported model selection strategies and performance interpretation (Raschka & Mirjalili, 2020).

IBM. (n.d.). *What is anomaly detection?* Retrieved from https://www.ibm.com/topics/anomaly-detection

This article helped in understanding the core concepts of anomaly detection (IBM, n.d.).
Scikit-learn Developers. (2023). *Scikit-learn User Guide*. Retrieved from https://scikit-learn.org/stable/user_guide.html

The official documentation was used as a technical reference for model implementation and parameter tuning (Scikit-learn Developers, 2023).

THE EXPERIENCES AND SKILLS ACQUIRED BY THE TEAM MEMBERS

Student Name	Experiences	Skills
Amal Abed Alobaidi	Participated in data preprocessing and text correction.	Data Cleaning, Python (pandas, re), Documentation
Khadijah Faletau	Led the classification analysis and handled model tuning.	Machine Learning (Classification), scikit-learn, F1 Score
Muhjah Mansha	Focused on anomaly detection and evaluated outlier models.	Anomaly Detection, Visualization (matplotlib, seaborn)
Rola Alsalem	Organized the dataset, corrected grades, and led regression analysis writing.	Regression Analysis, Report Writing, Data Analysis